

RESEARCH PLAN

In Vitro Cytotoxic Activity of Malunggay's (*Moringa oleifera*) Ethanolic Bark Extract against A549 Lung Cancer Cells

Proponents

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Rationale

Mitosis is the process by which eukaryotic cells divide to produce two genetically identical daughter cells. This process is important for understanding cell biology, but when mitosis malfunctions, it can result in a wide variety of health problems, with cancer being the most prominent (Jose et al., 2024). Cancer remains the second leading cause of death worldwide, accounting for approximately 10 million deaths globally, or about 1 in every 6 deaths. This is equivalent to roughly 19 deaths per minute (World Health Organization, 2022). It can appear in almost every part of the human body, such as the lungs, breasts, prostate, colon and rectum, liver, pancreas, kidneys, and stomach (National Cancer Institute, 2026). Due to the fact that some types of cancer progress silently without noticeable symptoms, early detection and treatment can be extremely difficult. That is why scientists continue to study its nature, and despite advances in technology and medicine, not all cancer subtypes currently have a definitive treatment (Saenz, 2024).

According to Mayo Clinic (2026), there are more than 100 types of cancer identified. Lung cancer is the most commonly diagnosed cancer and the leading cause of cancer mortality worldwide, with recent GLOBOCAN-based estimates reporting approximately 2.5 million new cases and 1.8 million deaths in 2022 (Bray et al., 2024; Zhou et al., 2024; Cancer Atlas, 2024). Major global risk factors identified by the World Health Organization (WHO) include tobacco use, ambient and household air pollution, occupational carcinogens, and second-hand smoke. In 2022, it accounted for approximately 2.5 million new cases, representing 11.5% of all cancer cases, and 1.8 million deaths, equivalent to 18.7% of all cancer-related deaths (Li et al., 2023). The disease remains disproportionately harmful to men, with incidence and mortality rates approximately twice as high as those among women. According to Li et al. (2023), the global burden of lung cancer is expected to continue increasing, particularly in low- and middle-Human Development Index countries where access to prevention, early detection, and treatment remains limited. In the Philippine context, lung cancer is the most common cancer among men, the second most common cancer overall, and the leading cause of cancer-related death in the country. According to the Department of Health (2025), the Philippines recorded approximately 23,728 new lung cancer cases and 20,953 deaths in 2024, accounting for nearly one out of every five cancer cases nationwide. These figures, together with the high prevalence of smoking and increasing exposure to other risk factors such as indoor air pollution and radon, emphasize the growing burden of lung cancer in the country and the urgent need to strengthen prevention, early detection, and treatment strategies.

Lung cancer treatment often involves surgical procedures such as lobectomy and pneumonectomy which, while effective, carry risks and complications including postoperative pulmonary complications such as atelectasis, pneumonia, prolonged air leak, and pleural

effusion, bleeding and hematoma, wound infection, arrhythmias particularly atrial fibrillation, respiratory failure, venous thromboembolism including deep vein thrombosis and pulmonary embolism, chronic postsurgical pain, reduced exercise tolerance, and in some cases, death (Sun et al., 2025).

On the other hand, other treatment options such as chemotherapy are also offered to patients. Chemotherapy uses drugs to stop or slow the growth of breast cancer cells. These drugs are typically administered intravenously through a vein via a needle or catheter or taken orally (Mayo Clinic, 2023). However, not all patients have access to these treatments. According to Booth et al. (2021), access to essential cancer medicines remains unequal worldwide, with many patients in most countries lacking access to basic cancer drugs.

Due to the limitations of conventional treatments, natural and plant-derived compounds have gained attention for their antiproliferative and pro-apoptotic properties. One of the most successful plant-derived anticancer drugs is vincristine (a vinca alkaloid) from the Madagascar periwinkle (*Catharanthus roseus*), which was the first plant-derived anticancer drug to enter clinical use and remains one of the most widely used chemotherapy agents worldwide (Abburi, 2025). Similarly, Korean red ginseng (*Panax ginseng*) has been found to be effective against MCF-7 breast cancer cells (Kim et al., 2021). Malunggay (*Moringa oleifera*) has also shown potential anti-cancer properties in different parts of the plant due to its bioactive chemical compounds.

Moringa oleifera, commonly known as Malunggay in the Philippines, is classified as an essential herbal plant due to its vast medicinal benefits and diverse pharmacological properties. (Banoub, 2026; Pareek et. al., 2023). Recent studies demonstrate that Malunggay offers antimicrobial, anticancer, anti-inflammatory, antidiabetic, and antioxidant benefits, expanding on

its traditional use for treating wounds, pain, ulcers, and liver disease. However, malunggay bark remains underutilized despite previous findings showing that *Moringa oleifera* leaf extracts possess strong anti-inflammatory, antioxidant, and anticancer properties (Abd-Rabou et al., 2023). Exploring this underutilized part of the plant can reveal additional therapeutic potential, particularly in developing natural, plant-based anticancer agents with unique bark-specific bioactive compounds. By evaluating how its ethanolic extract affects the viability of A549 lung cancer cells, this research could uncover novel insights, ultimately aiding the development of safer, more accessible, and plant-derived oncological treatments.

Given the complications associated with the current treatments of lung cancer, exploring safer alternatives is essential. This study aims to investigate the Cytotoxic Activity of *Moringa oleifera* bark ethanolic extract solution as a potential treatment for lung cancer. Using an experimental posttest-only control group design, A549 cells will be treated with varying concentrations of the peel solution and compared with Paclitaxel 3 through an MTT assay to assess cell viability, inhibition, and IC_{50} values. The results may contribute to the growing body of research on plant-based cancer therapeutics and help advance the development of less harmful treatment options

A. Problem Being Addressed

As lung cancer remains a global health concern, the A549 cell line is commonly used to evaluate the potential anticancer effects of various compounds. While conventional treatments are available, their significant risks and side effects highlight the need for alternative therapies. Malunggay (*Moringa oleifera*) bark may exhibit cytotoxic effects against lung cancer cells; however, its efficacy against A549 cells has not been fully investigated. Hence, this study seeks to answer the following research questions:

General Question

1. Can Malunggay's (*Moringa oleifera*) bark ethanolic extract exhibit in vitro cytotoxic activity against A549 lung cancer cells?

Specific Questions

1. What is the effect of Malunggay's (*Moringa oleifera*) bark ethanolic extract on the cell viability of A549 lung cancer cells in terms of:
 - 1.1. Dosage and;
 - 1.2. Comparative effectiveness with a known chemotherapeutic agent for (Doxorubicin)?
2. What is the effect of Malunggay's (*Moringa oleifera*) bark ethanolic extract on the cell inhibition of A549 lung cancer cells in terms of:
 - 2.1. Dosage and;
 - 2.2. Comparative effectiveness with a known chemotherapeutic agent (Doxorubicin)?
3. What is the effect of Malunggay's (*Moringa oleifera*) bark ethanolic extract on the IC₅₀ value of A549 lung cancer cells in terms of:
 - 3.1. Dosage and;
 - 3.2. Comparative effectiveness with a known chemotherapeutic agent (Doxorubicin)?

B. Expected Outcomes

This study is expected to demonstrate that the ethanolic extract exerts a dose-dependent cytotoxic effect on A549 lung cancer cells, as evidenced by reduced cell viability and increased

growth inhibition compared with the untreated control group in the MTT assay. It is anticipated that higher concentrations of the bark extract will yield lower cell viability percentages and a measurable IC₅₀ value, indicating its potential as an antiproliferative agent against lung cancer cells. When compared with Doxorubicin, the bark extract may show comparatively lower but still significant cytotoxicity, suggesting that while it may not fully match the potency of established drugs, it could serve as a complementary or alternative natural treatment with potentially fewer side effects and greater accessibility, especially in resource-limited settings like the Philippines. Overall, the expected outcomes may support the further exploration of *Moringa oleifera* bark as a promising source of plant-derived anticancer compounds, providing a scientific basis for its development into safer, more affordable breast cancer therapies.

C. Description in Detail of Methods and Procedure

C1. Procedures

The ethanolic extract of *Malunggay (Moringa oleifera)* bark will be prepared and utilized to evaluate its effects on cell viability, cell inhibition, and IC₅₀ values against A549 lung cancer cells. The procedures for extract preparation, experimental treatment, and MTT proliferation assay will be carried out following established laboratory protocols. Prior to the experiment, all required materials, reagents, and equipment will be prepared and organized to ensure the accuracy and consistency of the experimental procedures.

Research Design. This study will employ a true experimental posttest-only control group design to evaluate the potential anticancer activity of the ethanolic bark extract of *Moringa oleifera* against A549 lung cancer cells. The experimental groups will receive different concentrations of the extract, while the positive control group will be treated with doxorubicin, a commercially available chemotherapeutic agent, and the negative control group will receive

McCoy's medium. The independent variable will be the concentration of the malunggay ethanolic bark extract, whereas the dependent variables will include cell viability, cell inhibition, and IC₅₀ values. Data will be collected only after the treatments have been administered through the MTT proliferation assay. This posttest-only design will minimize potential pretesting effects and allow the observed outcomes to be attributed to the treatments applied.

Table 1

Treatment for Conditional Groups: Control and Experimental Setups

Setups	Dependent Variables		
	Cell Viability	Cell Inhibition	cValue
Control Setups		Treatments	
Negative Control		blank (McCoy 2)	
Positive Control		Doxorubicin	
Experimental Group	<i>Malunggay's Ethanolic Bark Extract</i>	Concentrations (ppm)	
Experimental Setup A	X_1	12.5	
Experimental Setup B	X_2	25	
Experimental Setup C	X_3	50	
Experimental Setup D	X_4	100	
Experimental Setup E	X_5	200	

Note. Adapted from “In Vitro Cytotoxic Activity of Watermelon (*Citrullus lanatus*) Peel Solution Against MCF-7 Breast Cancer Cells” by Apuli et al., 2026.

Table 1 presents the treatment design for the control and experimental groups that will be

used to evaluate the anticancer activity of *Malunggay's (Moringa oleifera) ethanolic bark extract* against A549 lung cancer cells. The table identifies the dependent variables, namely cell viability, cell inhibition, and IC₅₀ value, and outlines the corresponding control and experimental setups to be employed in the MTT proliferation assay.

The control groups will consist of a negative control and a positive control to establish baseline and reference responses. The negative control will contain McCoy's medium without treatment to represent normal cell behavior, while the positive control will be treated with doxorubicin, a standard chemotherapeutic agent, to serve as a reference for cytotoxic activity. These controls will provide benchmark values for evaluating the anticancer effects of the experimental treatments.

The experimental group will consist of five treatment setups (Experimental Setups A–E) containing different concentrations of *Malunggay's ethanolic bark extract*. Experimental Setup A (X₁) will receive 12.5 ppm, Setup B (X₂) will receive 25 ppm, Setup C (X₃) will receive 50 ppm, Setup D (X₄) will receive 100 ppm, and Setup E (X₅) will receive 200 ppm. These treatment concentrations will allow the researchers to assess the dose-dependent effects of the extract on cell viability and cell inhibition. In this study, the independent variable will be the concentration of the *Malunggay ethanolic bark extract*, whereas the dependent variables will be cell viability, cell inhibition, and IC₅₀ value. This experimental design will enable a systematic comparison between the treatment and control groups to determine the extract's potential cytotoxic and anticancer activities against A549 lung cancer cells.

Gathering and Preparation of Materials. The researcher will obtain 2kg of Malunggay's stem from one of the researcher's garden. This stem will be used in getting the bark for the

ethanolic extract needed for the study. It will then be sent to the Bureau of Plant Industry (BPI) for identification.

Production of Ethanolic Extract from Malunggay's Bark. As a general guideline from Pharmacognosy Department of the National University of Sadat City (2021) for medicinal plant processing, leaves, herbs, and flowers are typically dried at 40°C to 60°C, whereas barks and roots are often dried at 40°C to 80°C, depending on the plant material and the need to preserve active constituents. The Malunggay's (*Moringa oleifera*) bark was dried for 1 hour and 30 minutes, as adapted from the study of Apuli et al. (2026), in a 70°C oven and was ground into a powder using a food processor after the drying process. Adapted from the study of Al-Asmari et al. (2015), once the bark is dried, it will be ground into a coarse powder using a grinder. Their study used a 1:10 ratio adapted also from the study of R Hasibuan and E Gultom (2021) between the coarse powdered bark and ethanol respectively; the researchers will utilize that ratio using 25 grams of powder with 250mL of ethanol. The pulverized bark (25 grams) will then be placed inside a soft thimble (filter paper thimble), which will be inserted into the extraction chamber of the Soxhlet apparatus. A 70% ethanol solution (250 mL) will be used as adapted to the study of Ahmed et al. (2023) will be placed in the round-bottom flask of the Soxhlet apparatus and the extraction cycle will repeat continuously until the 3 cycle process is completed for 4-6 hours according to Ahamed et al (2026). This extraction procedure will be done 4 times to yield 1L of the ethanolic extract as recommended by the partner institution which is the Mindanao State University - Iligan Institute of Technology (MSU-IIT). After the Soxhlet extraction, the ethanolic extract will then be collected in the round-bottom flask and will be given to the partner institution. The institute will be the one to perform the rotary evaporator to remove the ethanol solvent under reduced pressure, leaving the crude *Moringa oleifera* bark extract for subsequent

laboratory analyses.

Observations and changes in the treated A549 lung cancer cells will be recorded throughout the experimental procedures. Data obtained from the MTT proliferation assay will be used to determine cell viability, cell inhibition, and IC₅₀ values. These parameters will be calculated using established mathematical formulas to evaluate the cytotoxic and antiproliferative effects of the ethanolic bark extract of *Moringa oleifera*. The results obtained from each treatment group will be compared with those of the positive control (doxorubicin) and the negative control to assess the extract's potential anticancer activity. Any significant differences among the treatment groups will be further evaluated using the appropriate statistical analyses.

Cell Viability. Cell viability measures the percentage of living cells within a specified cell population (Dewi et al., 2022). This parameter will be used to evaluate the cytotoxic effect of the extract on the treated cells. The positive control (doxorubicin) and the negative control (McCoy's medium) will serve as reference groups for comparison. Cell viability will be measured by determining the optical density (absorbance) at 570 nm. The percentage of cell viability will be calculated using the following formula:

$$\% Viability = \left[\frac{Absorbance_{sample} - Absorbance_{blank}}{Absorbance_{control} - Absorbance_{blank}} \right] \times 100$$

Cell Inhibition. Cell inhibition will be determined as the percentage of cells whose growth or survival is inhibited following treatment with the ethanolic bark extract of *Moringa oleifera*. This parameter will be used to evaluate the cytotoxic activity of the extract by measuring its ability to reduce the viability or cell death of A549 lung cancer cells. The

percentage of cell inhibition will be calculated using the following equation (Lintao & Medina, 2021):

$$\% \text{ Cytotoxicity} = 100 - \% \text{ Viability}$$

IC₅₀ Value. The half-maximal inhibitory concentration (IC₅₀) will be determined to evaluate the potency of the ethanolic bark extract of *Moringa oleifera* against A549 lung cancer cells. IC₅₀ represents the concentration of a substance required to inhibit 50% of cell viability and is widely used to assess the cytotoxic activity of potential anticancer agents. The IC₅₀ value will be calculated using linear regression analysis by plotting the logarithm of the extract concentration against the corresponding percentage of cell viability. A best-fit linear equation will then be generated, and the concentration corresponding to 50% cell viability will be determined using the equation of the line:

$$y = mx + b$$

In the equation, y represents the percentage of cell viability (dependent variable), m represents the slope of the best-fit line, x denotes the logarithm of the extract concentration (independent variable), and b represents the y-intercept.

C2. Risk and Safety

The risk is associated entirely with in vitro and does not include human or animal subjects, and includes the processing of plant tissues, ethanol, lab tools, and A549 lung cancer cells, every step will be done in a certified laboratory with qualified adviser supervising the work and following the usual biosafety plus lab safety protocols. Researchers will wear suitable personal protective equipment such as lab coats, gloves, face coverings, and safety glasses at all

times. Extra attention will be given while performing the extraction work with the oven, Soxhlet setup and rotary evaporator, mainly to avoid burns, unexpected leaks, or accidents tied to the equipment, and ethanol will be moved and kept properly because it is highly flammable.

C3. Data Analysis

Comparison of Experimental Set-up and Controlled Set-up. The collected data will be analyzed using Jamovi statistical software. Cell viability, cell inhibition, and IC₅₀ values will be expressed as mean ± standard deviation (SD). Prior to hypothesis testing, the normality of the data will be assessed using the Shapiro–Wilk test, with the test statistic calculated as follows:

$$W = \frac{(\sum_{i=1}^n a_i y_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where y_i represents the ordered sample values, $\bar{y}$ is the sample mean, a_i denotes the Shapiro–Wilk constants, and n is the total number of observations.

If the data are normally distributed, a one-way Analysis of Variance (ANOVA) will be performed to determine whether significant differences exist among the treatment groups. If the assumption of normality is not met, the Kruskal–Wallis test will be used as the non-parametric alternative. Statistical significance will be determined at $p < 0.05$.

The ANOVA F-ratio will be calculated using the following equation:

$$F = \frac{MST}{MSE}$$

Where, where F was the variance ratio for the overall test, MST represents the mean

square due to treatments (between-group variance) and MSE represents the mean square due to error (within-group variance).

The mean square due to treatments (MST) will be computed as:

$$MST = \frac{\sum_{i=1}^k (T_i^2 / n_i) - G^2 / n}{k - 1}$$

The mean square due to error (MSE) will be computed as:

$$MSE = \frac{\sum_{i=1}^k \sum_{j=1}^{n_i} Y_{ij}^2 - \sum_{i=1}^k (T_i^2 / n_i)}{n - k}$$

where Y_{ij} represents an individual observation, T_i is the total of observations in the i th group, G is the grand total of all observations, n_i is the number of observations in group i , k is the number of treatment groups, and n is the total number of observations. (Statsdirect, n.d.).

If the ANOVA indicates a statistically significant difference among the treatment groups, Tukey's Honest Significant Difference (HSD) test will be conducted as a post hoc analysis to identify which group means differ significantly. The HSD value will be calculated using the following equation:

$$HSD = q \sqrt{\frac{MS_{error}}{n}}$$

where HSD is the minimum significant difference between two group means, q is the critical value obtained from the Studentized range distribution, MS_{error} is the mean square error obtained from the ANOVA, and n is the number of observations per treatment group.

D. Cost Analysis

The cost analysis provides a comprehensive evaluation of the financial implications linked to the completion of the proposed research. This entails precisely identifying the expenses involved in obtaining materials, conducting essential tests, and producing the final research document. Through the analysis of costs, transparency and efficient budgeting is practiced among researchers, allowing for the strategic allocation of resources to support the study's objectives.

Table 1

Itemized Breakdown of the Costs of Materials, Procedures, and Research Documentation

Product Description	Quantity	Unit Price	Total Price
Airtight Container <i>from SM Store</i>	3	119.20	357.60
Filter Paper (24x24cm) <i>from Anscom Medical Center</i>	3	50	150
Petroleum Jelly (25g) <i>from Southstar Drug</i>	1	45	45
70% Ethanol <i>from Medic Diagnostic Reagents</i>	2	290	580
Acetone	2	10	20
Experiment Samples <i>from Partner Institutions</i>	3	5, 000	15, 000

Table 1 shows a comprehensive examination of the expenses associated with the

proposed research study, outlining the materials and tests to be conducted. Within each labeled entity, the table specifies the exact quantities required, the cost per unit for each item or service, the total cost, and the overall expenditure.

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